# Patentable Software-Only Invention Topics in Water, Electricity, and Traffic Analytics: A Landscape and Drafting Guide

## TL;DR

- **Yes, a purely software data-analytics method can be patented in India and the US — but only if the independent claim is framed around a concrete *technical effect* (a measurable improvement to a technical system), not an abstract calculation or business outcome. This single drafting choice is the difference between grant and rejection under India's Section 3(k) and the US Alice/Mayo test.**
- The safest targets are **narrow, sub-domain-specific methods in thinly-patented niches** — e.g., non-revenue water apportionment for *intermittent* supply, transformer-overload prediction from *billing-cycle* data, and accident-hotspot ranking from *government open crash records* — while broad areas (leak detection, load forecasting, congestion prediction, generic anomaly/theft detection) are heavily crowded and should be avoided as headline claims.
- No topic is guaranteed grantable; software analytics inherently invites Section 3(k) ("computer programme/algorithm per se") and Alice ("abstract idea") objections. Of the 10 topics proposed below, roughly the first four are the strongest, the middle three are moderate, and the last three are more marginal (higher prior-art or abstractness risk) — chosen because they are still excellent *academic* drafting exercises.

## Key Findings

**1. The legal bar is about framing, not subject matter.** Neither India nor the US bans software patents outright. India's Patents Act Section 3(k) excludes "a mathematical or business method or a computer programme *per se* or algorithms." The operative words are "per se" — the Delhi High Court in *Ferid Allani v. Union of India* (W.P.(C) 7/2014, decided 12 December 2019, Justice Prathiba M. Singh) held that a computer-related invention (CRI) demonstrating a "technical effect" or "technical contribution" is patentable even though based on a computer program, stating verbatim: "In today's

digital world, when most inventions are based on computer programs, it would be retrograde to argue that all such inventions would not be patentable. Innovation in the field of artificial intelligence, blockchain technologies and other digital products would be based on computer programs, however the same would not become non-patentable inventions – simply for that reason." CaseMine The US Alice/Mayo two-step test asks (1) is the claim "directed to" an abstract idea, and if so (2) does it recite an "inventive concept"/"significantly more" — and software that "improves computer functionality, solves a technical problem in an unconventional way, or produces a concrete technical result" Michaelmeyerlaw survives.

**2. "Business method" is an absolute bar in India — worse than a computer program.** In *OpenTV Inc. v. The Controller of Patents and Designs* (2023 SCC OnLine Del 2771; 2023:DHC:3305; Application No. 2564/DELNP/2012, decided 11 May 2023, Justice Prathiba M. Singh), the court upheld rejection of a "system and method to provide gift media," holding that "the bar in India to the grant of business method patents must be read as an absolute bar" Lexorbis and that the invention was "simply a method of doing business" — no technical-effect escape hatch. **Implication: any invention whose core is billing, pricing, tariff-setting, revenue-recovery, or customer-segmentation-for-marketing risks being killed as a business method.** These must be reframed around a downstream *technical* effect (network operation, data integrity, computational efficiency).

**3. Granted-vs-rejected Indian examples confirm the pattern.** In *Microsoft Technology Licensing v. Assistant Controller* (Delhi HC, 15 May 2023, 2023:DHC:3342), the court reversed a Section 3(k) rejection because the invention improved *network security* of user authentication — holding that "if the subject matter is implemented on a general-purpose computer, but results in a technical effect that improves the computer system's functionality and effectiveness, the claimed invention cannot be rejected … as 'computer program per se'," CaseMine and that requiring novel hardware "lacks any basis in law" (echoing *Raytheon*). In *BlackBerry* ("Auto-Selection of Media Files," 717/DEL/2009, C.A.(COMM.IPD-PAT) 318/2022), the Controller rejected the claim as a computer programme with no technical effect, but the Delhi High Court reversed because the invention "enhances the functionality and the capacity of the device," mirandah holding that "any invention which enhances the capability of a device beyond a mere algorithm or business method is not barred under Section 3(k)." mirandah The US contrast: in *Miller Mendel, Inc. v. City of Anna, Texas* (No. 2022-1753, 107 F.4th 1345, Fed. Cir., 18 July 2024), the Federal Circuit affirmed that claims of U.S. Patent No. 10,043,188 were ineligible under 35 U.S.C. §101, holding they were "directed to the abstract idea of performing a background check" IP UPDATE +2 and "merely recited well-understood, routine and conventional activities." IP UPDATE

**4. The 2025 CRI Guidelines raise the bar on disclosure.** India's Controller General

(CGPDTM) issued the final Guidelines for Examination of Computer Related Inventions on 29 July 2025 (replacing the 2017 version; preceded by draft v1.0 on 25 March 2025 and v2.0 on 26 June 2025). They codify a step-wise test requiring examiners to identify the core problem, map the solution, and determine whether the effect is "genuinely technical or merely functional and administrative," and demand that AI/ML claims not be "black boxes" — details and evidence are required, and system-level improvements (latency, throughput, memory) count.

**5. Prior-art density varies sharply by sub-area.** Based on the patent survey below, the areas are best classified as follows:

*Heavily crowded (AVOID as headline claim):*

- **Electricity theft / non-technical-loss detection from meter data** — extremely dense (e.g., US9600773B2 "Detecting electricity theft via meter tampering using statistical methods"; US20160161539A1; plus a large academic corpus).
- **Water leak detection from consumption/meter data** — dense (US10782204, US11359991, US11066813B2), though most tie to sensors/valves (hardware) — leaving a software-only gap.
- **Load forecasting & consumer segmentation from smart-meter data** — saturated in both patents and literature (clustering/k-means/fuzzy-c-means load profiling is textbook).
- **Traffic congestion / flow prediction from historical data** — dense (US9286793B2, US9996798, EP1528524B1, CN113178073A, CN113762338B).
- **Travel-time / route recommendation from historical congestion** — dense (US8537033B2, US9070291B2, US8548734B2, US20060004511A1).

*Thinner prior art (TARGET):*

- **Non-revenue water / apparent-loss apportionment purely from billing + DMA input records** — most patents are sensor/valve based; software-only water-balance methods are underrepresented in patents (mostly non-patent tools like AWWA Free Water Audit Software).
- **Transformer-overload prediction** — some patents exist (US10289954B2, US7701357B2, US7965193B2) but most rely on direct transformer power measurement (hardware); prediction purely from *consumer billing-cycle records mapped to transformers* is narrower.
- **Traffic data imputation for missing records** — active in academia but comparatively thin in granted patents (US11093463 is generic industrial imputation);

domain-specific traffic-count imputation methods are draftable.

- **Accident-hotspot identification from government crash open-data —** dominated by academic GIS/KDE methods and sensor-based systems (US12462673 uses road-user devices); a purely retrospective open-data method is less patented.
- **Billing-anomaly detection —** crowded generically (US20130339202A1, US20130030761A1, US11645727) but utility-specific, physically-constrained variants are draftable.

## Details

### How to make a data-analytics claim survive (the drafting playbook)

The recurring failure mode is a claim that reads: "receive data → compute a score/prediction → output a result." Under Alice step one this is "directed to" a mental process/mathematical concept; under Section 3(k) it is an "algorithm per se." The recurring *success* pattern has three ingredients:

1. **Anchor the claim to a technical problem in a physical/technical system**, not a commercial one. "Reducing false-positive alarms in a water distribution network's monitoring pipeline" is technical; "recovering lost revenue" is a business method (fatal in India per *OpenTV*).
2. **Recite a specific, unconventional technical step that produces the effect —** e.g., a particular data transformation, a constraint derived from physical laws (mass balance, Kirchhoff's current law), or a computational-efficiency improvement (reduced memory/compute for a given accuracy). Generic "apply machine learning" fails; a *specific* named transformation with a stated technical reason succeeds.
3. **Tie the output to a technical action or a measurable system-level improvement —** improved data integrity of a downstream control dataset, reduced computational load, improved accuracy quantified against a baseline. Under the 2025 CRI Guidelines you must *disclose and evidence* this effect, not merely assert it.

**Language that helps:** "technical effect," "improves the functioning of the [network/monitoring system]," "reduces computational/memory requirements," "improves the integrity/accuracy of the [downstream dataset]," physically-grounded constraints, and a concrete downstream technical action. **Language that hurts:** "business method," "revenue," "tariff/pricing optimization," "customer marketing/segmentation," "a user is presented with," and pure "calculate a score" claims with no technical anchor.

## The 10 proposed invention topics

For each: patent-style **Title**, **Problem**, **Novel/inventive step**, **Data source**, **Nearest prior art & distinction**, **Claim-framing tip**, **Dataset for feasibility**, and a **Strength rating**.

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**TOPIC 1 — STRONGEST Title:** *A Method for Apportioning Non-Revenue Water into Apparent and Real Loss Components in an Intermittently-Supplied Distribution Zone Using Only Billing and Bulk-Inflow Records* **Problem:** Standard water-audit tools (AWWA Free Water Audit Software) assume continuous (24×7) pressurized supply; they mis-estimate loss components under the *intermittent* supply typical of Indian cities, where supply-hours vary daily and meter under-registration behaves differently. **Novel/inventive step:** A supply-hours-normalized water-balance decomposition that corrects each meter's registration bias as a function of logged daily supply-duration, then attributes residual non-revenue water between apparent (metering/data-handling) and real (leakage) losses using a minimum-night-flow analogue reconstructed from billing cycles rather than sensors. **Data source:** District-metered-area (DMA) bulk-inflow totalizer readings + customer billing records + logged supply-hours. **Nearest prior art:** Sensor/valve-based NRW systems (Itron, Badger BEACON) and continuous-supply audit tools; distinction — no sensors, and an intermittency correction absent from continuous-supply models. **Claim-framing tip:** Frame as improving the *accuracy and integrity of the water-balance dataset used to operate the network* under intermittent supply — a technical effect on network operation, not "revenue recovery." **Dataset:** BWSSB ward-wise "Water Consumption Data" on data.gov.in; India-WRIS Water Audit module. **Why strong:** Intermittent-supply niche is genuinely under-patented and India-specific.

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**TOPIC 2 — STRONGEST Title:** *A Method for Predicting Distribution-Transformer Overload from Consumer Billing-Cycle Consumption Records Mapped to Transformer Service Areas* **Problem:** Utilities know transformer failures correlate with overload, but most prediction methods require transformer-mounted power sensors; DISCOMs often lack these on LT transformers yet possess consumer billing data. **Novel/inventive step:** Aggregating *consumer-to-transformer* mapped monthly billing records into a synthetic per-transformer load curve using a diversity-factor reconstruction, then predicting overload risk — deriving high-resolution risk from low-resolution billing data without any transformer instrumentation. **Data source:** Consumer billing records + consumer-transformer mapping (DISCOM GIS) + connected-load ratings. **Nearest prior art:** US10289954B2 (predicts from multi-source records but presumes richer feeder data);

US7701357B2/US7965193B2 (measure transformer power directly — hardware). Distinction — billing-only, no transformer measurement. **Claim-framing tip:** Technical effect = enabling overload prediction for *un-instrumented* assets, improving grid-monitoring dataset coverage. **Dataset:** CEEW smart-meter dataset (Mathura/Bareilly) on Kaggle; data.gov.in state-wise consumption. **Why strong:** The "billing-only, no sensor" constraint is a clean novelty and technical-effect hook.

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**TOPIC 3 — STRONGEST Title:** *A Method for Reconstructing Missing Traffic-Count Records in a Historical Volume Dataset Using Direction-of-Flow Conservation Constraints at Network Nodes* **Problem:** Open traffic-count datasets have gaps from logging failures; generic imputation ignores the physical constraint that vehicles entering a node must equal those leaving. **Novel/inventive step:** A flow-conservation-constrained imputation that enforces node-level inflow/outflow balance across a road-network graph as a hard constraint on the imputed values — improving imputation accuracy and internal consistency versus purely statistical/temporal methods. **Data source:** Public historical traffic-volume/count datasets with network topology. **Nearest prior art:** US11093463 (generic multiple imputation); academic DCNN-GAN, KNN, tensor methods. Distinction — the physical conservation constraint is a *technical* mechanism, not a statistical fit. **Claim-framing tip:** Technical effect = improved accuracy and physical consistency of the reconstructed dataset (quantify MAE reduction vs. baseline); emphasize computational method + constraint. **Dataset:** UCI traffic-flow-prediction datasets; OpenCity Bengaluru traffic datasets. **Why strong:** Physical constraint = strong Alice/3(k) technical anchor; imputation is thinner in patents than prediction.

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**TOPIC 4 — STRONG Title:** *A Method for Detecting Meter-Data Integrity Faults in Electricity Consumption Logs by Cross-Validating Aggregate Feeder Energy Against Summed Consumer Records* **Problem:** Distinguishing *data-quality faults* (missing/rolled-over/mis-keyed meter reads) from genuine consumption change is hard; treating them as theft produces false positives. **Novel/inventive step:** A reconciliation method that cross-checks summed consumer billing against feeder/DTR aggregate energy and classifies residual discrepancies into data-integrity fault categories (rollover, phase-swap, estimation-carryover) via physically-defined signatures — *before* any theft inference. **Data source:** Consumer meter logs + feeder/transformer aggregate energy + billing history. **Nearest prior art:** US9600773B2 and theft-detection corpus (jump straight to theft/anomaly). Distinction — this targets *data integrity of the metering dataset*, a technical effect, not fraud/revenue. **Claim-framing tip:** Frame as improving *integrity and reliability of the metering dataset* — explicitly a technical, not revenue,

effect (avoids business-method bar). **Dataset:** CEEW smart-meter dataset; NITI ICED AT&C-loss data. **Why strong:** Data-integrity framing sidesteps both the crowded theft space and the business-method trap.

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**TOPIC 5 — MODERATE Title:** *A Method for Ranking Accident-Prone Road Segments from Government Crash Records Using Exposure-Normalized Severity-Weighted Density Estimation* **Problem:** Raw crash-count hotspot maps over-flag high-traffic roads; they don't normalize by exposure or weight by severity consistently. **Novel/inventive step:** Combining severity-weighting with an exposure-normalization derived from registered-vehicle/road-length open data into a single network-kernel density estimator that ranks segments by *risk per unit exposure* rather than raw count. **Data source:** MoRTH / state crash records + registered-vehicle + road-length open data. **Nearest prior art:** Academic GIS/PKDE/network-KDE methods (widely published); US12462673 (uses live road-user devices — hardware). Distinction — retrospective, open-data-only, exposure-normalized. **Claim-framing tip:** The novelty-vs-academia risk is real; emphasize the *specific* exposure-normalization computation and a technical effect on prioritization-dataset accuracy. Note honestly: heavy academic prior art. **Dataset:** MoRTH "Road Accidents in India" (OpenCity CSV mirror); Kaggle India Road Accident datasets. **Why moderate:** Rich academic literature makes inventive-step harder; still a clean software-only exercise.

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**TOPIC 6 — MODERATE Title:** *A Method for Estimating Apparent Water Losses Attributable to Meter Under-Registration by Classifying Consumption-Decay Signatures in Long-Term Billing Histories* **Problem:** Aging mechanical water meters progressively under-register; utilities cannot easily separate genuine reduced consumption from meter degradation using billing data alone. **Novel/inventive step:** A per-meter longitudinal signature classifier that distinguishes gradual metrological decay (a characteristic slow monotonic decline independent of seasonality) from behavioral consumption change, quantifying apparent loss per meter. **Data source:** Multi-year customer billing/meter-reading histories. **Nearest prior art:** NRW software (component analysis via lab meter-testing samples); leak-detection patents. Distinction — infers meter degradation from billing time-series, no field testing/sensors. **Claim-framing tip:** Technical effect = improved accuracy of the apparent-loss estimate in the water-balance dataset; avoid "revenue." **Dataset:** BWSSB ward-wise consumption; synthetic degradation on UCI/water datasets. **Why moderate:** Novelty plausible but overlaps NRW apparent-loss literature.

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**TOPIC 7 — MODERATE Title:** *A Method for Imputing Missing Intervals in Electricity Load Records Using Same-Category Consumer Cohort Substitution Constrained by Billing Totals* **Problem:** Smart-meter/DISCOM interval data has gaps; naive imputation breaks the constraint that imputed intervals must sum to the known billed total for the cycle. **Novel/inventive step:** Cohort-based imputation (borrowing shape from similar-tariff-category consumers) with a hard rescaling constraint so imputed intervals reconcile exactly to the billed monthly total — guaranteeing dataset consistency. **Data source:** Interval load records + monthly billing totals + tariff category. **Nearest prior art:** Generic imputation (US11093463) and load-profiling literature. Distinction — the billing-total reconciliation constraint. **Claim-framing tip:** Technical effect = reconciled, internally-consistent load dataset; quantify accuracy improvement. **Dataset:** CEEW smart-meter dataset; Kaggle India electricity consumption. **Why moderate:** Constraint gives a technical hook, but imputation broadly is active art.

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**TOPIC 8 — MODERATE-MARGINAL Title:** *A Method for Deriving Green-Time Split Recommendations at a Signalized Intersection from Historical Turning-Movement Count Records Using Saturation-Flow-Normalized Demand Ratios* **Problem:** Signal timing is often outdated; recomputing splits normally needs live detectors. **Novel/inventive step:** Computing phase demand ratios normalized by saturation flow purely from *historical* turning-movement counts to output offline timing-plan splits, with a fairness/starvation constraint — no live detection. **Data source:** Historical turning-movement/vehicle-count datasets + intersection geometry. **Nearest prior art:** US10559201, EP3158547A1, US20080204277A1 (all use live probe/connected-vehicle/detector data). Distinction — historical-record-only, offline. **Claim-framing tip:** Technical effect = improved timing-plan (reduced modeled delay); tie to a concrete computed control parameter, not "traffic management" generally. **Dataset:** Bengaluru City Traffic Signal Data (OpenCity); UCI traffic datasets. **Why marginal:** Signal-optimization space is crowded; the historical-only limitation is the main novelty lever.

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**TOPIC 9 — MARGINAL Title:** *A Method for Detecting Systematic Billing-Estimation Errors in Utility Consumption Records by Identifying Rounding and Estimation-Carryover Signatures Across Consumer Cohorts* **Problem:** "Estimated" (non-actual) meter reads propagate errors across cycles; detecting systematic estimation bias in historical billing is hard. **Novel/inventive step:** A cohort-level detector for the characteristic saw-tooth signature of estimated-then-trued-up reads, flagging records whose pattern indicates estimation carryover rather than real consumption. **Data source:** Historical billing records with actual/estimated read flags. **Nearest prior art:** Generic billing-anomaly patents (US20130339202A1, US20130030761A1,

US11645727). Distinction — specific to the estimation-carryover physical artifact. **Claim-framing tip:** Frame as *data-integrity* of the consumption dataset, NOT billing correctness (business-method risk in India is high here). **Dataset:** CEEW smart-meter data; BWSSB consumption. **Why marginal:** Closest to the business-method line; must be carefully technicalized.

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**TOPIC 10 — MARGINAL Title:** *A Method for Classifying Recurrent vs. Non-Recurrent Congestion on a Road Segment from Historical Speed Records Using Day-Type Consensus Profiles* **Problem:** Distinguishing predictable (recurrent, e.g., rush-hour) from event-driven (non-recurrent) congestion in historical data helps planning but is usually done with live incident feeds. **Novel/inventive step:** Building per-segment "consensus day-type" baseline profiles and classifying each historical interval as recurrent/non-recurrent by deviation from its own consensus, purely retrospectively. **Data source:** Historical segment speed/flow datasets. **Nearest prior art:** Congestion-prediction patents (US9286793B2, EP1528524B1) and "consensual days" academic work. Distinction — classification (not prediction), offline, no incident feed. **Claim-framing tip:** Technical effect on the *derived classification dataset*'s accuracy; quantify vs. baseline. **Dataset:** UCI traffic-flow datasets; OpenCity Bengaluru. **Why marginal:** Congestion analytics is heavily patented and academically saturated; hardest inventive-step argument.

### Public datasets (consolidated)

- **Electricity:** CEEW smart-meter dataset (Mathura/Bareilly, 3-min interval, 2019–2021) on Kaggle; "Electricity Consumption: Year-, Month- and State-wise" (Dataful/Ministry of Power); NITI Aayog ICED dashboard (AT&C losses, billing/collection efficiency); data.gov.in Power sector; AIKosh/IndiaAI.
- **Water:** BWSSB ward-wise "Water Consumption Data" (data.gov.in); "Households access to safe drinking water" (data.gov.in); India-WRIS / NWIC (groundwater + Water Audit module); UCI Water Treatment Plant dataset (id 106).
- **Traffic:** MoRTH "Road Accidents in India" (OpenCity CSV mirror, 2019–2023; data.gov.in 2019/2020); Bengaluru Traffic Police datasets + Bengaluru City Traffic Signal Data (OpenCity); "Total Number of Registered Motor Vehicles in India" (data.gov.in); UCI traffic-flow-prediction datasets; Kaggle India Road Accident datasets; Smart Cities Mission portal.

## Recommendations

**Stage 1 — Pick from the strong tier first.** For a 30-mark assignment, choose Topics

1–4. They combine (a) a clean software-only architecture, (b) a physical/technical constraint that gives a genuine Section 3(k)/Alice technical-effect anchor, and (c) thinner prior art. Topic 3 (flow-conservation traffic imputation) and Topic 2 (billing-only transformer overload) are the two best single choices because the physical constraint doubles as both the inventive step *and* the technical effect.

**Stage 2 — Draft the independent claim to the playbook.** Structure every independent claim as: [receive specified data] → [a *specific named* technical transformation, stated with its physical/technical rationale] → [apply the physically-grounded constraint] → [produce a dataset/parameter whose technical improvement is measurable] → [a downstream technical action or quantified system-level improvement]. Add dependent claims narrowing the constraint and the data types.

**Stage 3 — Pre-empt the two killer objections in the specification.** (i) Include a "Technical Effect" paragraph explicitly naming the improvement (accuracy %, computational/memory saving, dataset-integrity gain) with evidence from your prototype — the 2025 CRI Guidelines demand disclosed, evidenced effects. (ii) Scrub all business-method language (revenue, tariff, pricing, marketing) — *OpenTV* makes these an absolute bar in India.

**Benchmarks that change the recommendation:** If a prior-art search (Google Patents + espacenet + Indian patent search) turns up a granted claim within one narrowing limitation of your chosen topic, drop to the next topic in the strong/moderate tier. If you cannot articulate a *measurable* technical effect for a topic, treat it as marginal and reframe or replace it.

## Caveats

- **No grant is guaranteed.** Software-analytics claims are among the hardest to prosecute; both Section 3(k) and Alice are applied unpredictably, and examiner-level outcomes vary. The Indian patent system is dominated by large and often foreign applicants: per ORF (citing WIPO), India ranked sixth globally in patent grants, issuing 76,053 patents in 2023, of which 77.7% went to non-residents; in 2024–25 only about 10,682 of 33,504 grants (~32%) went to Indian applicants (Scroll.in / Economic Survey 2025–26) — reflecting how much the outcome depends on sophisticated drafting.
- **Academic prior art counts as prior art.** Many of these areas (especially Topics 5, 8, 10) have dense peer-reviewed literature. A patent needs novelty *and* non-obviousness over *all* public disclosures, not just patents. The assignment likely rewards a defensible *argument*, but be honest in the write-up about nearest academic work.

- **This is guidance, not legal advice.** Verify each topic with a fresh prior-art search before drafting; patent landscapes shift, and the 2025 CRI Guidelines were finalized only on 29 July 2025, so examination practice around them is still settling.
- **India vs. US divergence:** A claim tuned for India's "technical effect" test generally also helps under Alice, but the business-method absolute bar is stricter in India — optimize for the Indian test since that is the student's likely jurisdiction.